

TASK-2 REPORT

Vulnerability Scanning & Service Enumeration Using Nmap

Student Details

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- **Course:** B.Tech CSE (Cyber Security)
- **Task:** Task-2
- **Internship:** Ethical Hacking & Penetration Testing

1. Objective

The objective of this task is to perform vulnerability scanning and service enumeration on the Metasploitable2 virtual machine using Kali Linux and Nmap. The task helps in identifying open ports, active services, operating system details, and potential vulnerabilities in a controlled penetration testing environment.

2. Vulnerability Scanning Concepts

Vulnerability scanning is the process of identifying weaknesses, open ports, and insecure services in a system.

Key Points:

- Helps identify security weaknesses
- Detects open ports and services
- Supports penetration testing activities
- Improves system security

3. Service Enumeration

Service enumeration is the process of gathering detailed information about services running on a target system.

Enumeration Includes:

- Port identification
- Service detection, Version detection
- Operating system detection

4. Lab Setup

A virtual lab environment was created using VMware Workstation.

Tools Used:

- Kali Linux (Attacker Machine)
- Metasploitable2 (Target Machine)
- VMware Workstation
- Nmap

Network Configuration:

- Network Type: Host-only (VMnet1)
- IP Range: 192.168.199.xxx

5. Connectivity Testing

The connection between Kali Linux and Metasploitable2 was verified using ping.

Command:

```
ping 192.168.199.129
```

Result:

Successful communication was established between both virtual machines.

6. Advanced Nmap Scan

An advanced Nmap scan was performed to identify ports, services, operating system details, and vulnerabilities.

Command:

```
nmap -sS -sV -O -A 192.168.199.129
```

Observation:

The scan successfully identified multiple open ports, running services, and operating system information on the target machine.

7. Full Port Scan

A full TCP port scan was performed to detect all open ports on the target machine.

Command:

```
nmap -p- 192.168.199.129
```

Observation:

Several open ports and vulnerable services were identified during the scan.

8. Open Ports and Services Identified

Port	Service
21	FTP
22	SSH
23	Telnet
25	SMTP
80	HTTP
139	NetBIOS
445	SMB
3306	MySQL
5432	PostgreSQL
5900	VNC
6667	IRC
8180	Apache Tomcat

9. Vulnerabilities Identified

The following vulnerabilities and insecure services were identified:

- Anonymous FTP login enabled
- Telnet service active
- SMB service exposed
- Old Apache server version detected
- Multiple unnecessary open ports
- Outdated Linux kernel version

10. Analysis

The scanning results showed that the Metasploitable2 machine contains multiple intentionally vulnerable services. Nmap successfully detected open ports, service versions, and operating system details. This task improved practical understanding of reconnaissance and vulnerability assessment techniques.

11. Challenges Faced

- Slow scanning due to multiple open ports
- Some services taking more time to respond
- Large scan output difficult to analyze initially

Solutions:

- Used aggressive scanning options
- Verified connectivity before scanning
- Carefully analyzed Nmap results

12. Conclusion

This task provided practical knowledge of vulnerability scanning and service enumeration using Nmap. Multiple open ports and vulnerable services were identified successfully on the Metasploitable2 machine. The task improved understanding of reconnaissance, network scanning, and vulnerability assessment in a secure virtual lab environment.

13. Screenshots / Evidence

Figure 1: Lab Setup

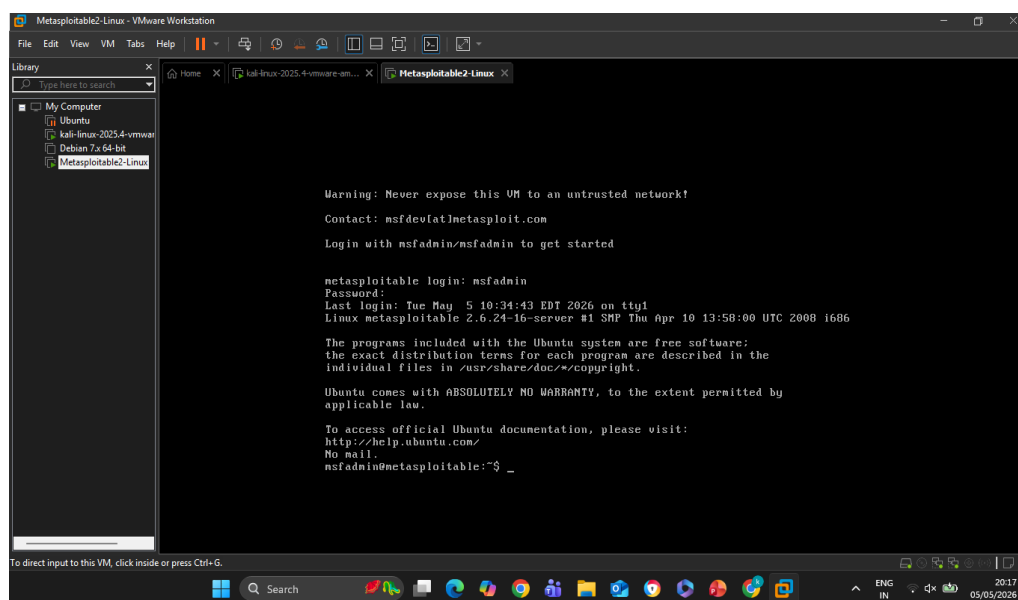
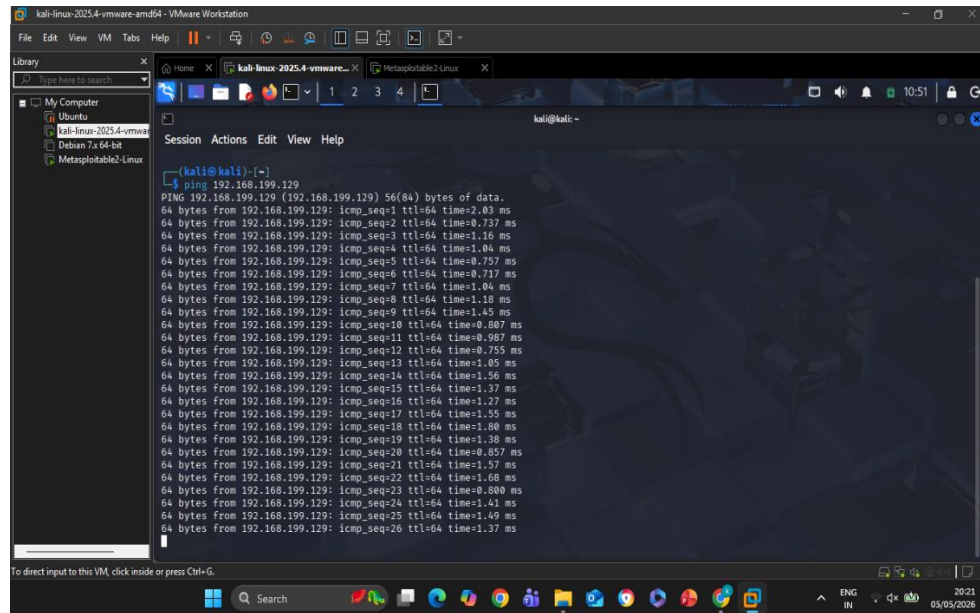


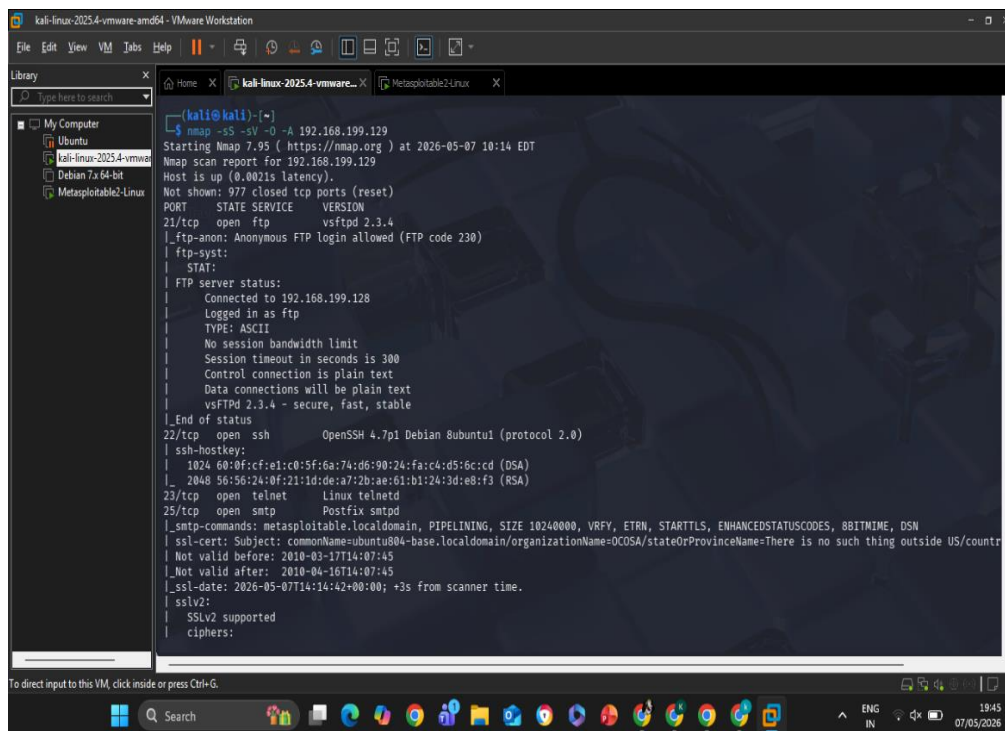
Figure 2: Ping Result



The screenshot shows a Kali Linux terminal window with the command `ping 192.168.199.129` executed. The output displays 26 successful ping requests, each receiving 64 bytes of data from the target IP. The response times are consistently low, ranging from approximately 0.737 ms to 1.8 ms. The terminal window is titled "kali@kali: ~" and the background features a Kali Linux logo.

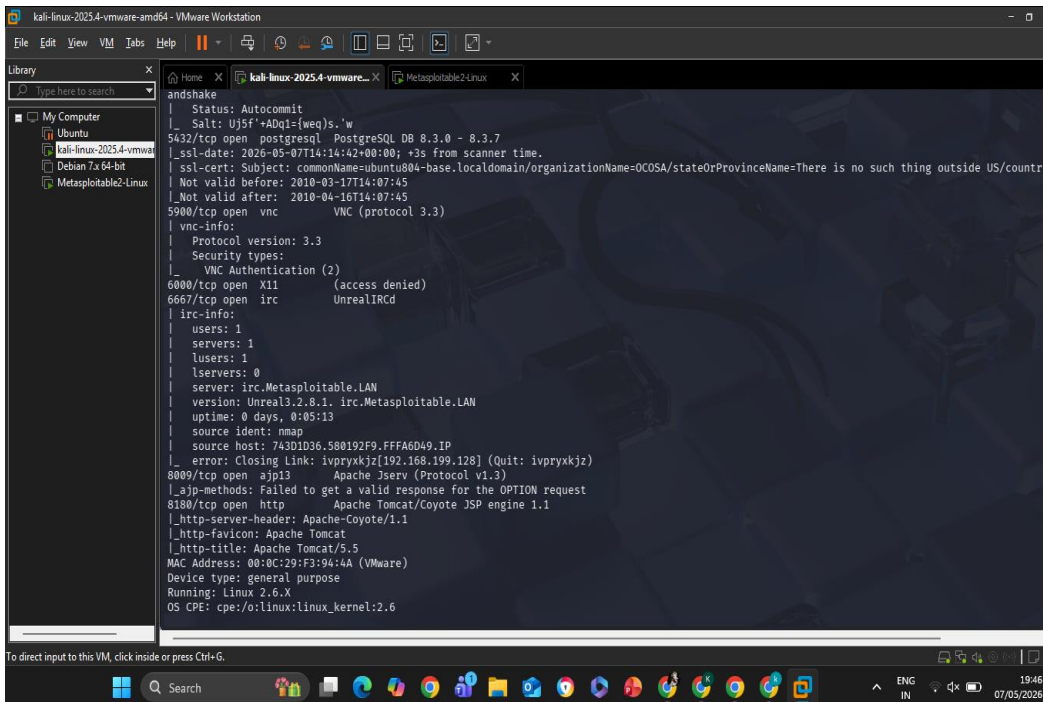
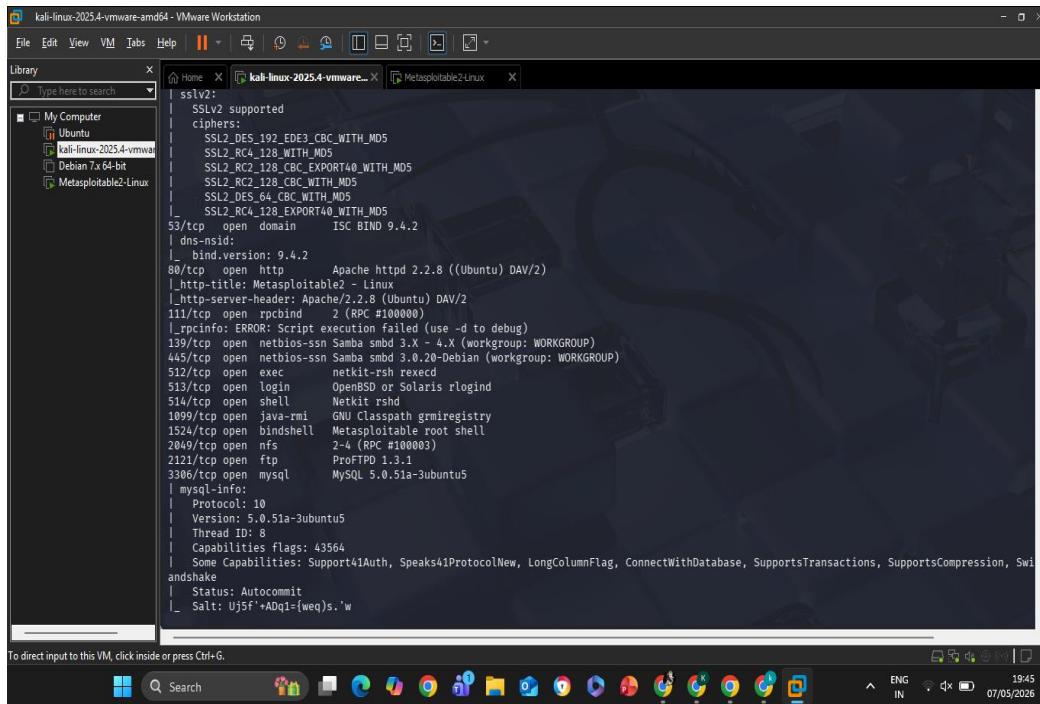
```
kali@kali: ~  
$ ping 192.168.199.129  
PING 192.168.199.129 (192.168.199.129) 56(84) bytes of data:  
64 bytes from 192.168.199.129: icmp_seq=1 ttl=64 time=2.03 ms  
64 bytes from 192.168.199.129: icmp_seq=2 ttl=64 time=0.737 ms  
64 bytes from 192.168.199.129: icmp_seq=3 ttl=64 time=1.16 ms  
64 bytes from 192.168.199.129: icmp_seq=4 ttl=64 time=1.04 ms  
64 bytes from 192.168.199.129: icmp_seq=5 ttl=64 time=0.757 ms  
64 bytes from 192.168.199.129: icmp_seq=6 ttl=64 time=0.717 ms  
64 bytes from 192.168.199.129: icmp_seq=7 ttl=64 time=1.04 ms  
64 bytes from 192.168.199.129: icmp_seq=8 ttl=64 time=1.18 ms  
64 bytes from 192.168.199.129: icmp_seq=9 ttl=64 time=1.45 ms  
64 bytes from 192.168.199.129: icmp_seq=10 ttl=64 time=0.807 ms  
64 bytes from 192.168.199.129: icmp_seq=11 ttl=64 time=0.987 ms  
64 bytes from 192.168.199.129: icmp_seq=12 ttl=64 time=0.755 ms  
64 bytes from 192.168.199.129: icmp_seq=13 ttl=64 time=1.05 ms  
64 bytes from 192.168.199.129: icmp_seq=14 ttl=64 time=1.56 ms  
64 bytes from 192.168.199.129: icmp_seq=15 ttl=64 time=1.37 ms  
64 bytes from 192.168.199.129: icmp_seq=16 ttl=64 time=1.27 ms  
64 bytes from 192.168.199.129: icmp_seq=17 ttl=64 time=1.55 ms  
64 bytes from 192.168.199.129: icmp_seq=18 ttl=64 time=1.88 ms  
64 bytes from 192.168.199.129: icmp_seq=19 ttl=64 time=1.38 ms  
64 bytes from 192.168.199.129: icmp_seq=20 ttl=64 time=0.857 ms  
64 bytes from 192.168.199.129: icmp_seq=21 ttl=64 time=1.57 ms  
64 bytes from 192.168.199.129: icmp_seq=22 ttl=64 time=1.68 ms  
64 bytes from 192.168.199.129: icmp_seq=23 ttl=64 time=0.880 ms  
64 bytes from 192.168.199.129: icmp_seq=24 ttl=64 time=1.41 ms  
64 bytes from 192.168.199.129: icmp_seq=25 ttl=64 time=1.49 ms  
64 bytes from 192.168.199.129: icmp_seq=26 ttl=64 time=1.37 ms
```

Figure 3: Advanced Nmap Scan



The screenshot shows a Kali Linux terminal window with the command `nmap -sS -sV -oA 192.168.199.129` executed. The output provides a detailed report of the scan, including host information, open ports, and service versions. The scan identifies several open ports: 21/tcp (vsftpd), 22/tcp (ssh), 23/tcp (telnet), and 25/tcp (smtp). The terminal window is titled "kali@kali: ~" and the background features a Kali Linux logo.

```
kali@kali: ~  
$ nmap -sS -sV -oA 192.168.199.129  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-05-07 10:14 EDT  
Nmap scan report for 192.168.199.129  
Host is up (0.0021s latency).  
Not shown: 977 closed tcp ports (reset)  
PORT      STATE SERVICE      VERSION  
21/tcp    open  ftp          vsftpd 2.3.4  
|_ftp-anon: Anonymous FTP login allowed (FTP code 230)  
|_ftp-syst:  
|_STAT:  
|_FTP server status:  
|_  Connected to 192.168.199.128  
|_  Logged in as ftp  
|_  TYPE: ASCII  
|_  No session bandwidth limit  
|_  Session timeout in seconds is 300  
|_  Control connection is plain text  
|_  Data connections will be plain text  
|_vsFTPd 2.3.4 - secure, fast, stable  
|_End of status  
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)  
|_ssh-hostkey:  
|_ 1024 60:0f:cfe1:c0:5f:6a:74:d6:90:24:fa:c4:d5:6c:cd (DSA)  
|_ 2048 56:56:24:0f:21:1d:de:a7:2b:ae:61:b1:24:3d:e8:f3 (RSA)  
23/tcp    open  telnet       Linux telnetd  
25/tcp    open  smtp         Postfix smtpd  
|_smtp-command: metasplitable.localdomain, PIPELINING, SIZE 10240000, VRFY, ETRN, STARTTLS, ENHANCEDSTATUSCODES, 8BITMIME, DSN  
|_ssl-cert: Subject: commonName=ubuntu0804-base.localdomain/organizationName=OCOSA/stateOrProvinceName=There is no such thing outside US/countryName=  
|_Not valid before: 2010-03-17T14:07:45  
|_Not valid after: 2010-04-16T14:07:45  
|_ssl-date: 2026-05-07T14:14:42+00:00; +3s from scanner time.  
|_sslv2:  
|_SSLv2 supported  
|_ciphers:
```



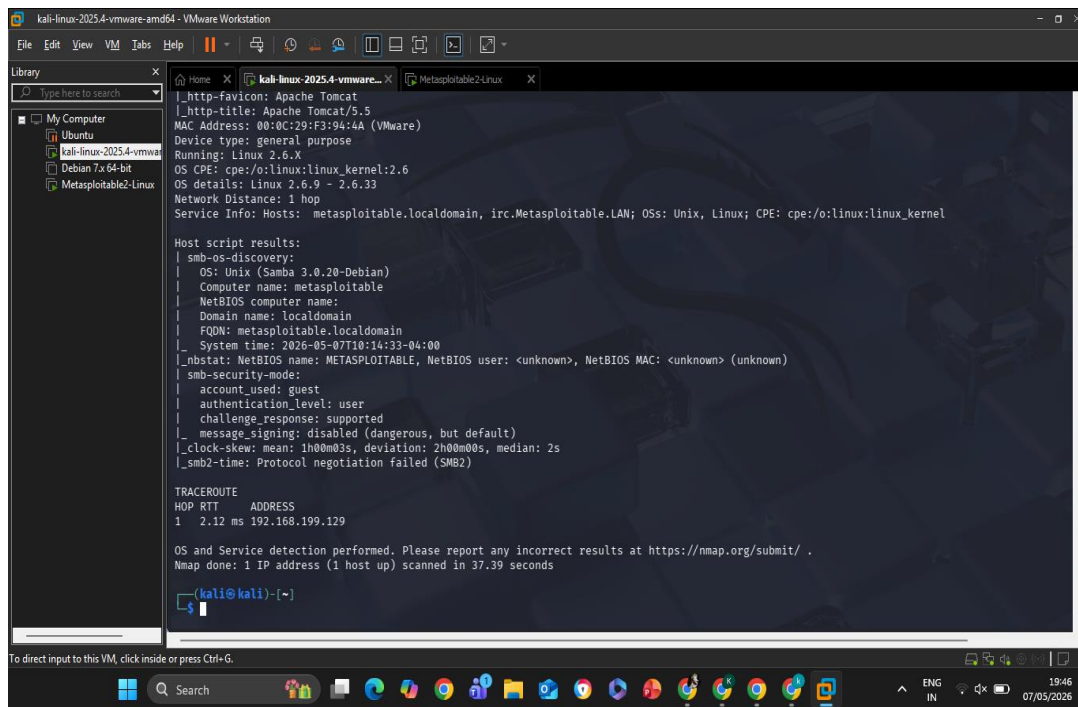
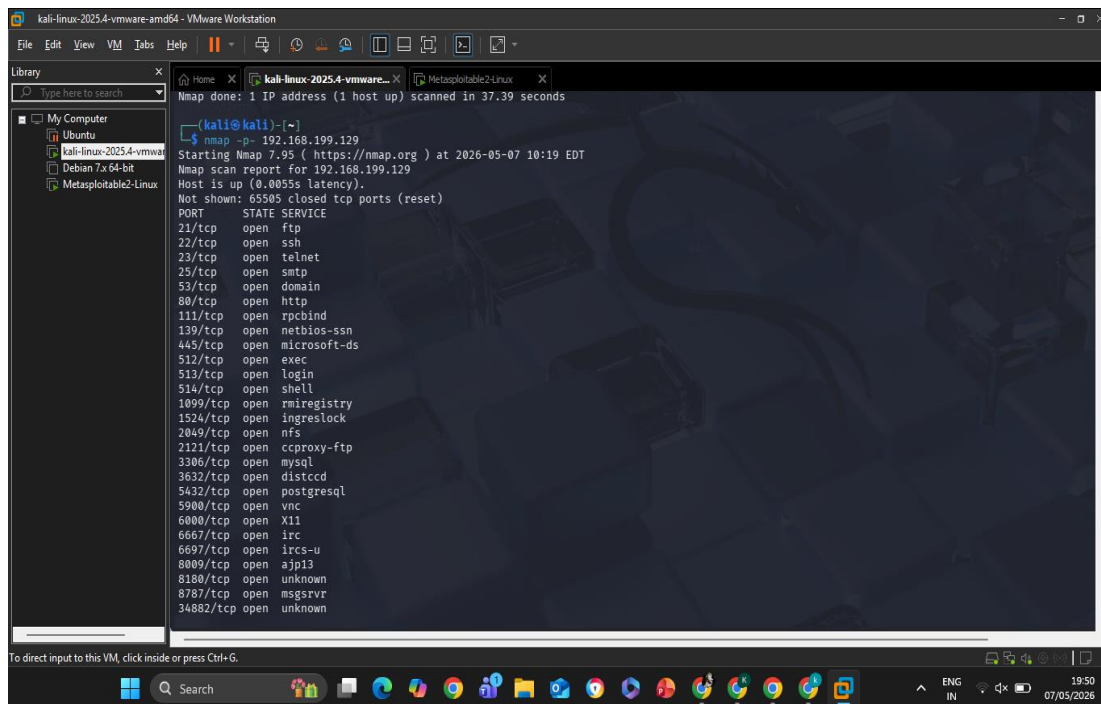


Figure 4: Open Ports & Services



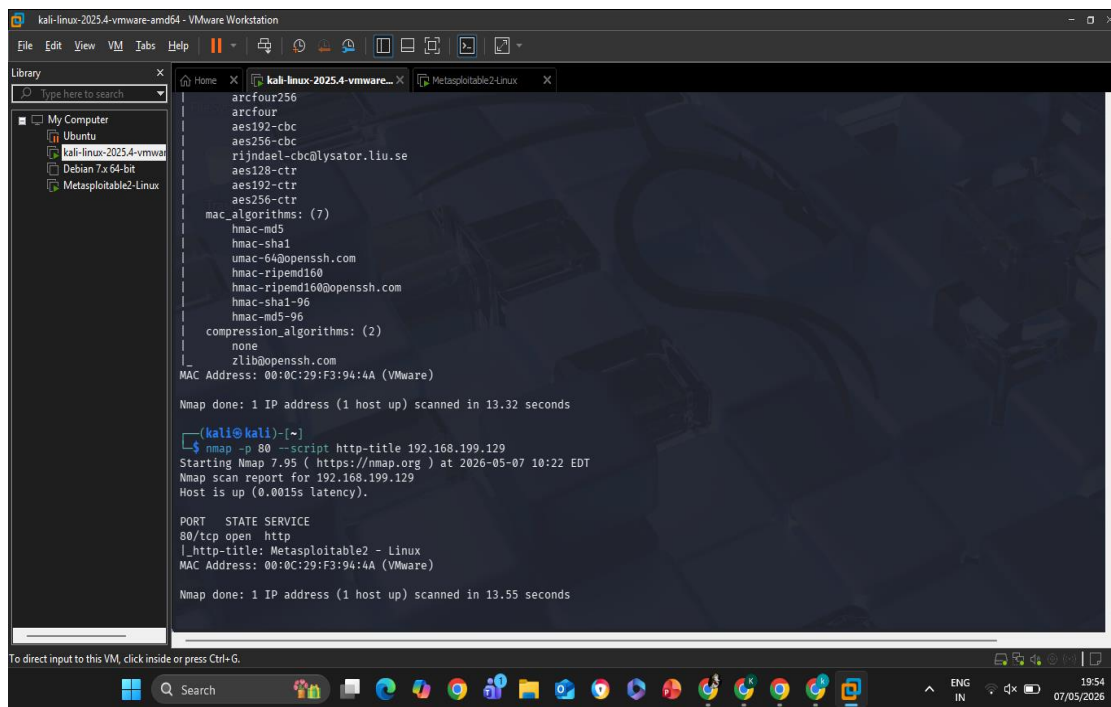
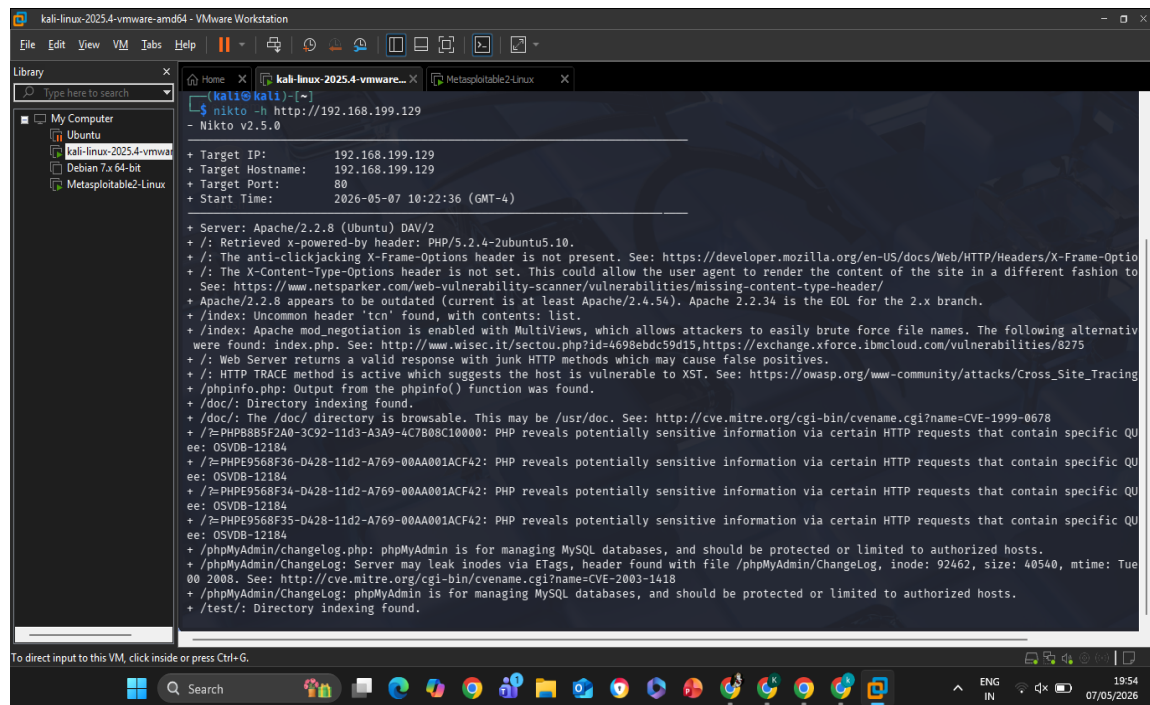
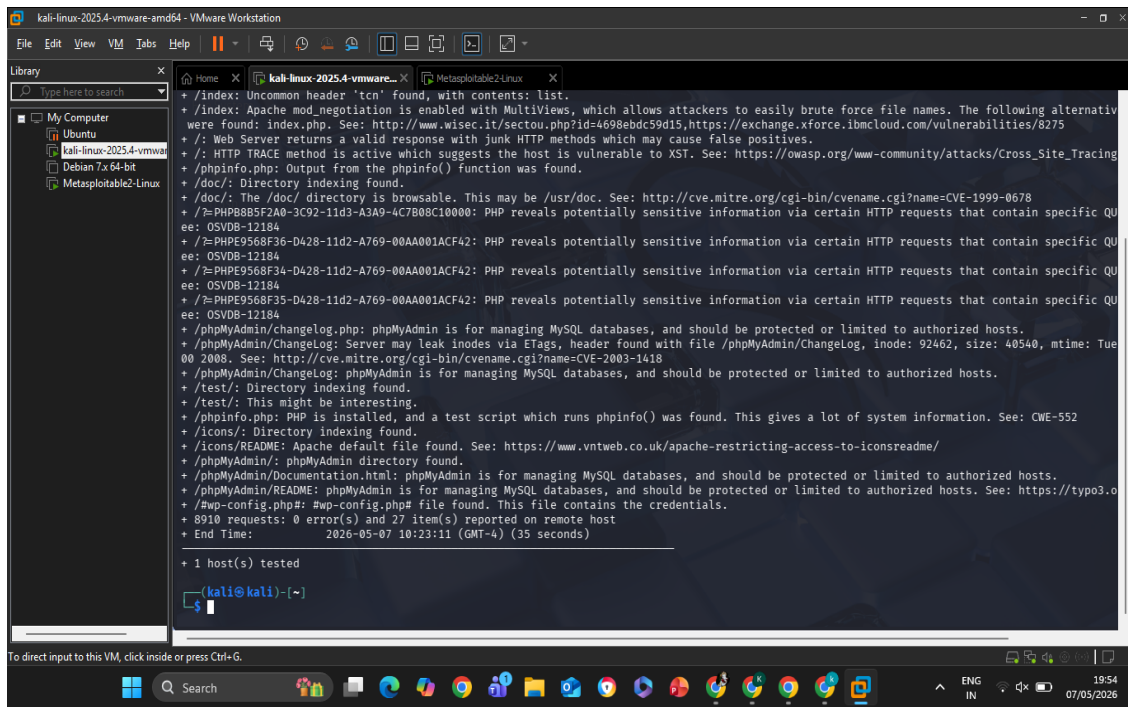


Figure 6: Nikto Web Vulnerability Scan





End of Report